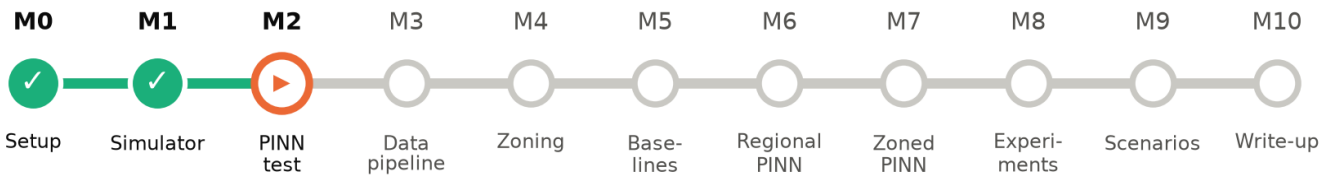
Zoned PINNs for Groundwater Forecasting

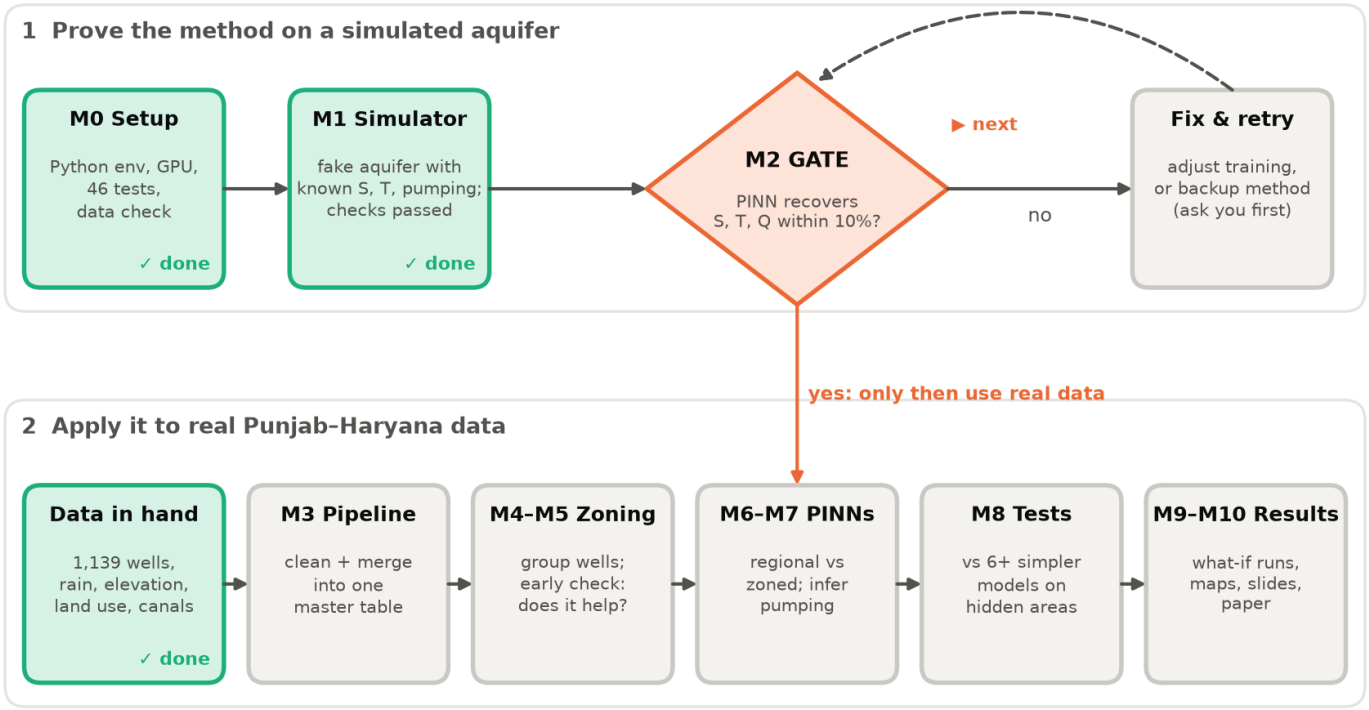
Progress report · 17 October 2026 · Punjab–Haryana alluvial belt, 16 districts · **2 of 11 milestones complete**



The project in one paragraph

We forecast groundwater levels everywhere in the study area, including places with no monitoring well. Wells are grouped into **zones** that behave alike. Each zone gets a **physics-informed neural network (PINN)**: it fits the measured water levels while obeying the groundwater flow equation **$S \cdot \partial h/\partial t = T\nabla^2 h + R - Q$** (storage × change in level = sideways flow + recharge − pumping). Unmetered farm **pumping Q** is inferred as a hidden quantity. Before touching real data, the method must prove on a simulated aquifer that it recovers answers we planted.

Project flow and where we are



Done so far

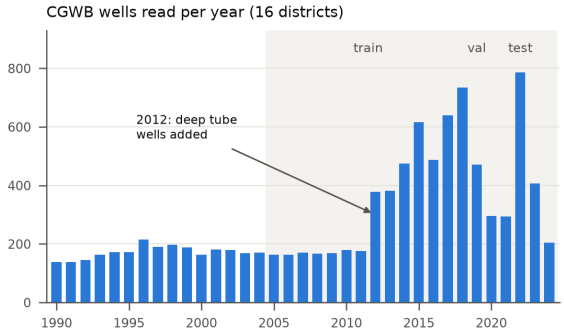
Step	What was done	Key result
Plan	Brief checked; 12 design problems ranked and fixed (PLAN.md)	Pumping made identifiable by tying it to land use and the crop calendar
Data	Study area chosen by a data check; every dataset fetched by script	1,139 wells (1990–2024) + 248 sensor wells; nothing left to download by hand
M0 Setup	Python environment, project layout, automatic tests, Colab notebook, literature check	GPU (GTX 1050) works; 46 tests pass; no prior zoned or hidden-pumping PINN on field data found
M1 Simulator	Finite-difference aquifer simulator and 4 synthetic aquifers (A–D) with known answers	Matches exact textbook solutions; water balance closes to 1 part in 10 billion

Biggest risks found in the plan review, and how they are handled

Risk	Why it matters	How the plan handles it
Pumping can't be pinned down	Many mixes of storage, recharge and pumping fit the same water levels	Pumping tied to land use and crop season; tested on simulated data first; reported as ranges
Physics may add little	Wells are 5–10 km apart, but water moves only ~1 km sideways per season	A simple 'bucket' model and a no-sideways-flow version are tested, so we can say so honestly
Result could be luck	A small test set makes a 5% difference look like chance	859 test wells; 3 different held-out area layouts; confidence ranges; main result fixed in advance
PINN training fails	PINNs often fail to train on messy, sparse data	Careful scaling, staged training, automatic tests; a backup method if needed (asks first)

Data in hand

Dataset	Coverage
CGWB water levels (India-WRIS)	1,139 wells, 1990–2024, with well type and depth
Sensor wells (6-hourly)	248 wells, mostly 2022–2025
IMD rainfall (0.25°, daily)	All 35 years, 1990–2024
Elevation (Copernicus 30 m)	Whole belt
Land use (Bhuvan, ESA)	2005-06, 2011-12, 2015-16, 2021
Canals, districts, crop areas	OSM canals; crop areas 1997–2025
CGWB reports (PDF)	Aquifer maps for all 16 districts; assessments 2020–24

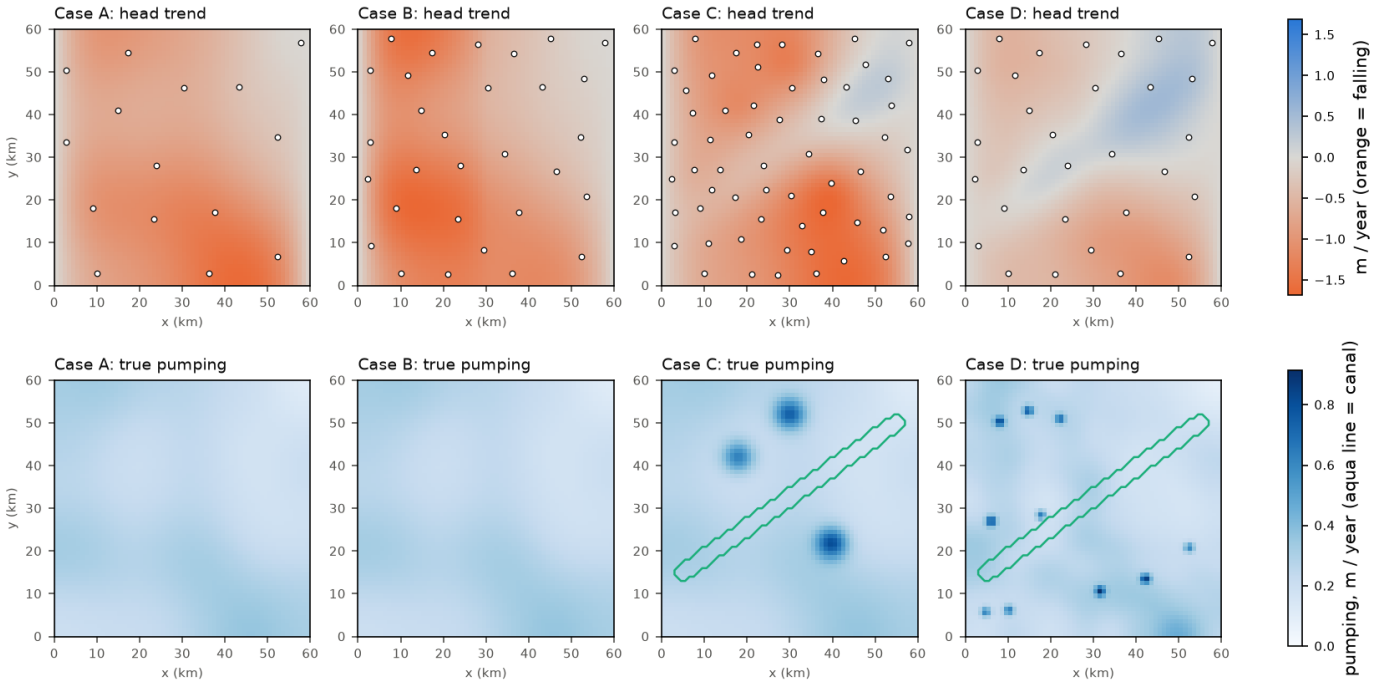


Usable for the main analysis: 1,031 wells (shallow, unconfined aquifer) · **193** with long enough records to define zones · **859** wells in the 2020–24 test years (663 with earlier history, 196 brand new: a fair test of predicting where the model has never seen data).
Watch-outs: the network shifted from shallow dug wells to deeper tube wells in 2012 (check that zones are not just well types); Karnal is thin in WRIS (filled from the India Data Portal); land use exists as snapshots, not every year.

M1 result: the simulated aquifer is verified

Check against an exact answer	Required	Result
Grid made 2× finer → error shrinks	≈ 4×	3.9× ✓
Water conserved every simulated day	< 1 in 10^8	1 in 10^{10} ✓
Closed box / straight-line cases	exact	exact ✓
Single pumping well (Theis)	close	within 0.4% ✓

Four test aquifers (60 × 60 km, 10 years):
A uniform, falls ~0.9 m/yr: main test
B two zones with different S, T: can zoning find them?
C canal + 3 heavy-pumping spots (4–7 m dips): **the M2 pass/fail test**
D deliberately unfair: hidden recharge, patchy pumping, blurry land-use map: measures how wrong the model gets when its assumptions are wrong



Top: 10-year water-level trend in each test aquifer (orange = falling; dots = the sparse 'wells' the PINN will see). Bottom: the true pumping the PINN must recover without being shown it (aqua outline = canal).

Next steps and honest status

Next	What it decides
M2 PINN on the simulated aquifer	Pass/fail gate: recover S, T and pumping within ~10% on case C. If it fails, fix or switch to the backup method (asks first). No real-data modelling before this passes.
M3 data pipeline	Clean and merge all data into one table; convert depth-to-water into water level.
M4–M5 zoning + early check	First real signal on whether zoning beats a single regional model.

Honest status: no accuracy result on real data exists yet. The 5–10% improvement over a single regional PINN is a *target, not a result*; the first evidence comes from the zoning check in M4. Pumping estimates on real data will be inferences with no ground truth and will be reported as ranges.